

Coding & Solution

Date	03 October 2026
Team ID	SWTID-2026-9455
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

The complete source code, configuration files and setup instructions are in the repository root (app/, templates/, static/, tests/, scripts/, requirements.txt, .env.example, Dockerfile, README.md).

Solution Summary

Field	Details
Repository Link / URL	https://github.com/keerthivasanB2007/ComicCraft
Programming Language(s)	Python 3.10+, HTML5, CSS3, JavaScript (small), Jinja2 templating
Framework(s) Used	FastAPI, Uvicorn, Pydantic, Jinja2; Google Gen AI SDK (google-genai); Hugging Face Diffusers + PyTorch; FPDF2; Pillow; pytest
Key Features Implemented	Story input form; input validation; 5-panel outline (Gemini Flash); narration and dialogue (Gemini Pro); panel illustrations (Stable Diffusion with placeholder fallback); layout builder; timestamped PDF export; comic preview, download and success pages; JSON API; /test-image; /health; offline mock mode; 21 automated tests; performance test script; Dockerfile
Pending / Incomplete Features	None for the planned scope. Accounts, comic library, character consistency and cloud hosting are future scope.
Setup / Run Instructions	1) python -m venv env and activate it. 2) pip install -r requirements.txt. 3) Copy .env.example to .env and add GEMINI_API_KEY and HF_API_KEY. 4) uvicorn app.main:app --reload. 5) Open http://127.0.0.1:8000 (API docs at /docs). No keys? Set COMICCRAFT MOCK=1. Tests: pytest -q

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organised into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes (verified by the team with real Gemini and Stable Diffusion on 30 Sep 2026; 21/21 automated tests pass)
6	Code is committed to a version control repository	Yes - pushed to the team GitHub repository (link on the Project Executable Files page)

Additional Notes / Comments

- The brief names **gemini-1.5-flash**, **gemini-1.5-pro** and **runwayml/stable-diffusion-v1-5**. Google has retired the Gemini 1.5 models and the Stable Diffusion 1.5 weights moved to stable-diffusion-v1-5/stable-diffusion-v1-5, so all three model IDs are set in .env (GEMINI_FLASH_MODEL, GEMINI_PRO_MODEL, SD_MODEL_ID). The models

actually used and tested are **gemini-2.5-flash** (outline) and **gemini-3.5-flash** (story); gemini-2.5-pro was listed but returned 404 for our free API key and gemini-3.1-pro had no free quota. The two-step design (outline, then story) is unchanged.

- The current google-genai SDK is used because the older google-generativeai SDK is deprecated.
- Stable Diffusion inference is serialised with a lock because diffusers pipelines are not thread-safe.
- Template names follow the milestone document (index.html, comic_preview.html, export_success.html).